

Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io

1. Input AC supply (low frequency, high voltage AC)

Mains power enters — in most countries this is 50/60 Hz AC at 110–240V, shown as the sine wave on the far left.

2. Rectifier (first one)

A bridge rectifier (four diodes) converts the AC sine wave into pulsating DC — the waveform changes from a smooth sine to a series of humps **high voltage unfiltered DC**

3. Filter (first one)

A capacitor smooths those humps into a relatively **flat High voltage DC**.

4. High-frequency switch (MOSFET)

This is the heart of an SMPS. A transistor (usually a MOSFET) rapidly switches the DC on and off at a high frequency (tens of kHz to MHz), turning smooth DC into a **high-frequency High voltage square wave AC**.

5. Step-down transformer

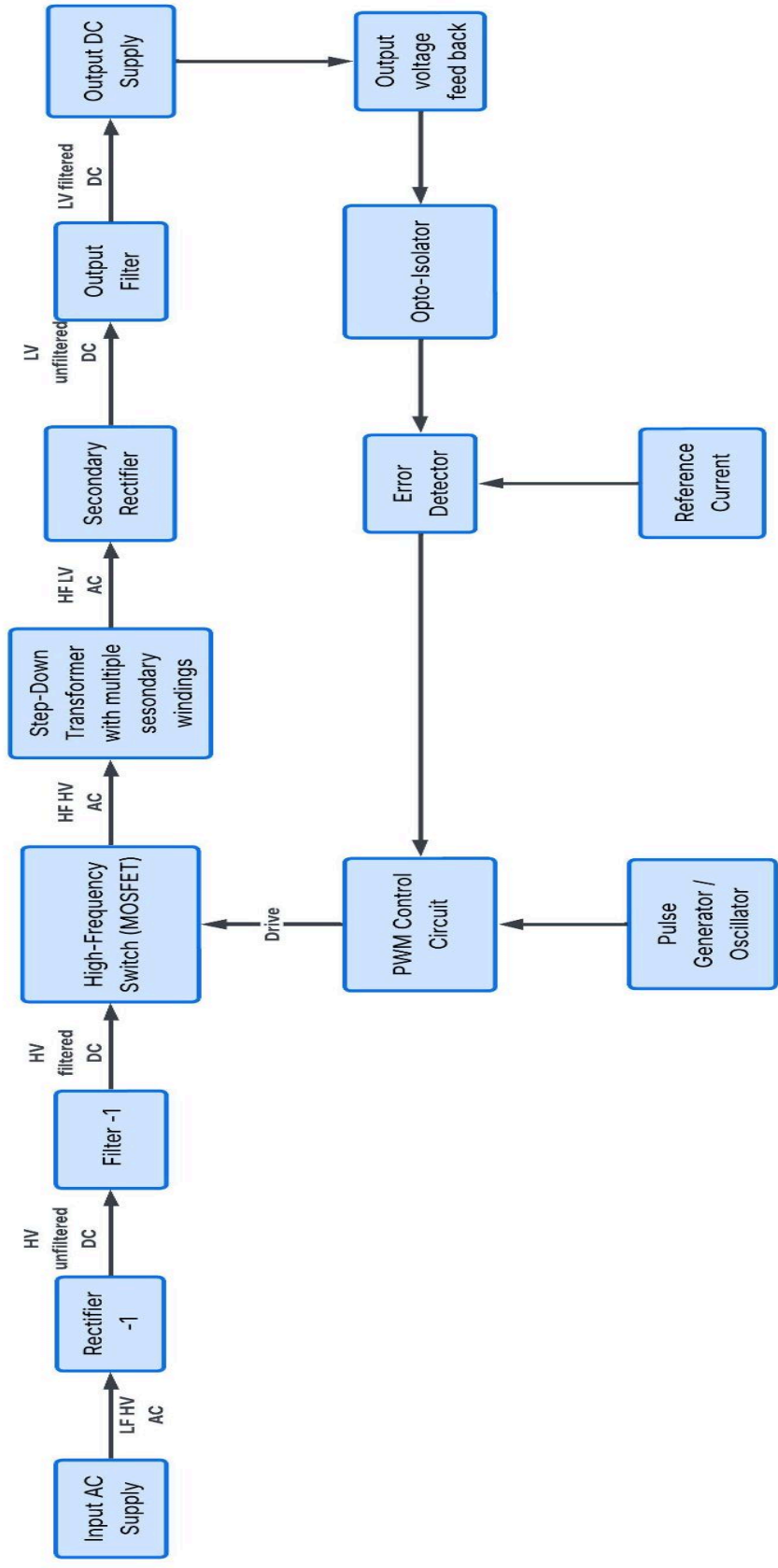
The high-frequency square wave passes through a small, light transformer that steps the voltage down (and provides electrical isolation between input and output for safety). Output is now **high frequency, low voltage AC**.

6. Rectifier (second one)

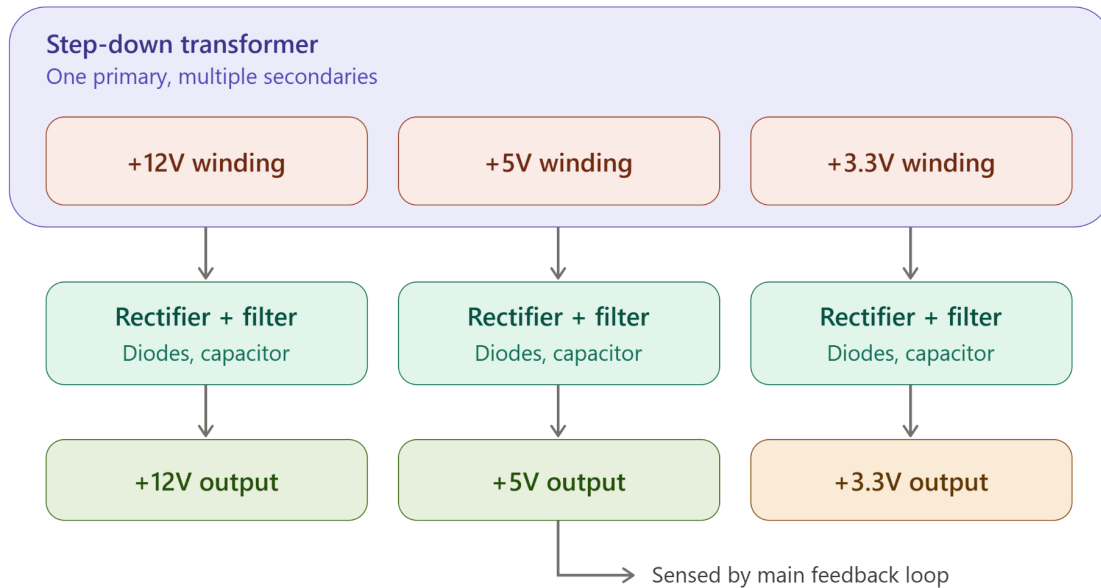
A second rectifier converts this low-voltage high-frequency AC back into **pulsating DC**.

7. Filter (second one)

A final filter smooths it into clean, **steady low-voltage DC** — the usable output that powers your device.



Note: the Step down transformer has multiple secondary windings, thereby providing different voltages. Each winding gets its own rectifier + filter. Only one rail usually drives the feedback loop



The +12V and +3.3V rails follow the +5V rail's switching automatically (same primary switching), with small secondary regulators trimming each one to stay accurate.

The feedback/control loop — this is what makes it "switched mode":

A small sensor checks what voltage is actually coming out, and that reading gets compared against a fixed reference (like "this should be +5V"). If the output is a bit low or a bit high, that's the *error*. This error signal tells the PWM (**pulse-width modulation**) **control circuit** whether to leave the switch on a little longer (boosts voltage) or cut it off sooner (drops voltage) on each cycle.

The high-frequency switching transistor's *on/off ratio* (duty cycle) gets operated accordingly. More "on" time per cycle → more energy passed through → higher output. Less "on" time → lower output.